DTC INSPECTION [EPS CONTROL MODULE]

id060230813000

1. Connect the M-MDS to the DLC-2.
2. After the vehicle is identified, select the following items from the initialization screen of the M-MDS.
   (1) Select “Self Test”.
   (2) Select “All CMDTCs”.
3. Verify the DTC according to the directions on the screen.
   - If any DTCs are displayed, perform troubleshooting according to the corresponding DTC inspection. (See DTC TABLE [EPS CONTROL MODULE].)

   Note
   - Snapshot data appears at the top of the help screen when the displayed DTC is selected.
   - Snapshot data stores the currently detected DTC data.

4. After completion of repairs, clear all DTCs stored in the module. (See CLEARING DTC [EPS CONTROL MODULE].)

Snapshot Data Table

Note
- Refer to PID monitor table for confirm the EPS system operation status while the EPS does not store the DTC. (See PID/DATA MONITOR TABLE [EPS CONTROL MODULE].)
- Snapshot data items are not displayed, according to detected DTC.

—: Not applicable

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
AAT	°C, °F	Ambient air temperature	—	—
APP_STATUS	Accelerator Pedal Off/ Under 20%/ Over 20%/ FAIL	Accelerator pedal position	—	—
CEN_TRQ_S	Nm	Center value of torque sensor	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error	Configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0 - Under 80 degrees C/ Over 80 degrees C/ FAIL	Engine coolant temperature status	—	—
ECU_IN_TMP	°C, °F	ECU internal temperature	—	ECU_IN_TMP
ENG_RPM	RPM	Engine speed	—	ENG_RPM
IC_VPWR	V	Instrument cluster power supply	• The EPS control module constantly receives the power supply voltage value of the instrument cluster sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the M-MDS.	—

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
IG-ON_TIMER	hh:mm:ss*1	Elapsed time since ignition was switched ON **Note** • The instrument cluster records the elapsed time since the ignition was switched ON.	• The EPS control module constantly receives the elapsed time since the ignition was switched ON sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the elapsed time since the ignition was switched ON when the DTC was detected, and it is displayed in the M-MDS.	—
MT_CURRENT	A	EPS motor control current	—	MT_CURRENT
OH_CR_C	No/Yes	Current overheat protection control (complete)	—	OH_CR_C
OH_CR_M	No/Yes	Current overheat protection control (middle)	—	OH_CR_M
OH_HIST_C	No/Yes	History of overheat protection control (complete)	—	OH_HIST_C
OH_HIST_M	No/Yes	History of overheat protection control (middle)	—	OH_HIST_M
OH_IG_CNT_C	—	The number of times the ignition is switched ON during the period from when overheat protection control (complete) is finished to the time the next overheat protection control (complete) is started • Displays a maximum number of 255 times in which the ignition is switched ON until the first overheat protection control (complete) is started. • During overheat protection control (complete), the number of times the ignition is switched ON is not counted, and 0 is displayed.	—	OH_IG_CNT_C

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
OH_IG_CNT_M	—	The number of times the ignition is switched ON during the period from when overheat protection control (middle) is finished to the time the next overheat protection control (middle) is started • Displays a maximum number of 255 times in which the ignition is switched ON until the first overheat protection control (middle) is started. • During overheat protection control (middle), the number of times the ignition is switched ON is not counted, and 0 is displayed.	—	OH_IG_CNT_M
PWR_MODE_KEY	Key Out/Key Recently Out/Key Approved (Position 0)/Post Accessory (Position 0)/ Accessory (Position 1)/ Post Ignition (Position 1)/Ignition On (Position 2)/Running (Position 2)/ Running - Starting In Progress (Position 2)/ Crank (Position 3)	• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time within 3 s since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time within 3 s since ignition was switched ON • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting: Cranking condition	• The EPS control module constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	—
RPM_STATUS	Engine Stop/ Under 1500 rpm/ Over 1500 rpm/ FAIL	Engine RPM status	• The EPS control module constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	—

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
SHIFT_STATUS	P/N D/ R/ FAIL	Shift position status	• The EPS control module constantly receives the selector lever position sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the selector lever position when the DTC was detected, and it is displayed in the M-MDS.	—
STR_ANG	°	Steering wheel angle	—	STR_ANG
STR_ANG_EST	°	Steering wheel angle (estimated absolute angle)	—	STR_AB_EST
STR_ROT_SPD	°/s	Steering wheel rotation speed	—	STR_ROT_SPD
STR_TRQ_S_M	Nm	Steering shaft torque (Main)	—	STR_TRQ_S_M
STR_TRQ_S_S	Nm	Steering shaft torque (Sub)	—	STR_TRQ_S_S
TOTAL_DIST	km, ft, mi	Accumulated total traveled distance from completion of vehicle until EPS control module detects DTC (Odometer value in instrument cluster)	The distance traveled when the EPS control module detected a DTC can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snap shot data item TOTAL_DIST. 3. Subtract 2 from 1.	—
TOTAL_TIME	hh:mm:ss*1	Accumulated total elapsed time since vehicle completion until EPS control module detects a DTC **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time when the EPS control module detected a DTC can be calculated by performing the following procedure. 1. Verify the PID item TOTAL_TIME of the instrument cluster. 2. Verify the snap shot data item TOTAL_TIME. 3. Subtract 2 from 1.	—
VPWR	V	Power supply	—	VPWR
VSPD	KPH, MPH	Vehicle speed	—	VSPD

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
VSPD_STATUS	Stop/ 0 - 10 km/h/ Over 10 km/h/ FAIL	Vehicle speed status	• The EPS control module constantly receives the vehicle speed sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the vehicle speed when the DTC was detected, and it is displayed in the M-MDS.	—

*1 : Fractions of seconds may be indicated after the decimal point